

DEUTSCHE BANK
AI IN FINTECH BUILDATHON 2024

TRACK 1- MARKET DATA RISK

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LINK FOR GITHUB
REPOSITORY

THE PROBLEM STATEMENT

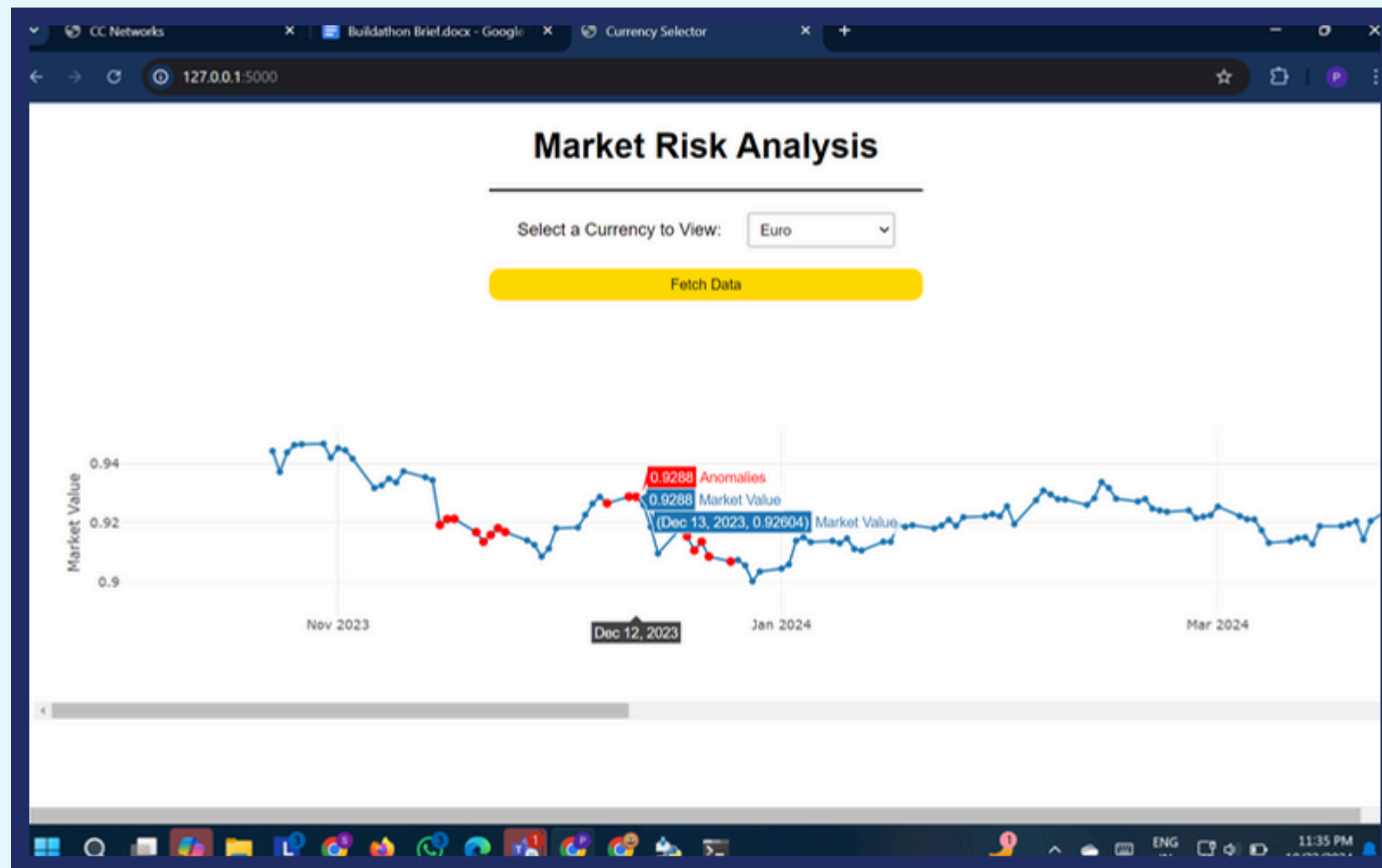


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Market Data Risk

- **Region Risk:** Analyze and mitigate risks arising from market data anomalies on a global scale, ensuring data integrity and reliability
 - **Challenge Clue 1:** Implement machine learning models that can predict and classify market anomalies in real-time and create a dashboard that visualizes these anomalies and their potential impacts.
 - **Challenge Clue 2:** Develop a system that can automatically flag and explain significant deviations in market data, providing actionable insights for risk mitigation.
- **Currency Risk:** Focus on managing risks associated with currency fluctuations. Create models to predict currency movements and develop strategies to hedge against potential losses.
 - **Challenge Clue 1:** Design a simulation tool that can test various hedging strategies under different market scenarios and provide recommendations based on historical data and predictive analytics.
 - **Challenge Clue 2:** Build an alert system that notifies users of significant currency movements and suggests immediate actions to mitigate risks.

REGION RISK ANALYSIS DASHBOARD



- **Technologies Used**
 - Flask: Framework for building the web application.
 - Plotly: Library for interactive data visualization.

Key Features

- **Currency Selection:**
 - Choose a specific currency to analyze.
 - Each currency represents its respective region (e.g., EUR for Europe, USD for the US).
- **Real-Time Market Data:**
 - Fetches historical market prices using the yfinance library.
 - Provides up-to-date information for selected currencies and cryptocurrencies.
- **Anomaly Detection:**
 - Anomalies are detected using a 1.5 standard deviation threshold from the rolling standard deviation of closing prices.
 - Anomalies are visually highlighted on the Plotly graph, allowing for easy identification of significant market deviations.

News Related to Anomalies

- **News Retrieval:**

- For each identified anomaly, news articles are fetched based on the corresponding date.
- News from one day prior to the anomaly date is considered for context.

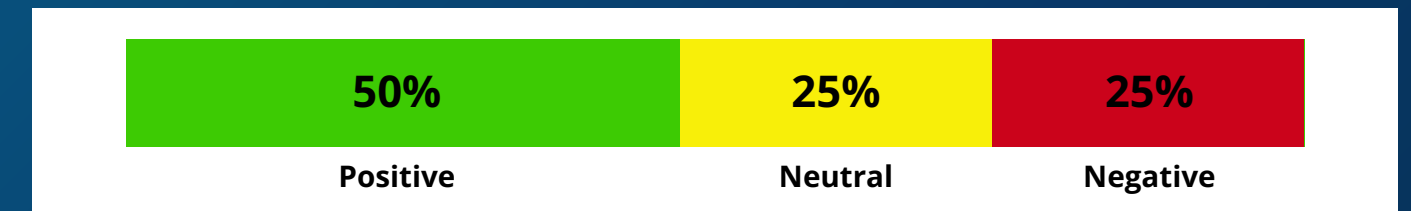
- **News Summarization:**

- Utilizes a pre-trained transformer model to summarize headlines.
- Generates concise summaries of the most relevant news articles for each anomaly date.

Sentiment Analysis

- **Sentiment Estimation:**

- Employs the FinBERT model to assess the sentiment of news headlines.
- Classifies sentiments into three categories: Positive, Negative, and Neutral.



- Helps in understanding market sentiment around anomalies, providing context for market movements.

- **Future Predictions**

- Market Direction Prediction:
- Based on the sentiment analysis of news articles, predicts potential market movements.
- Estimates the likelihood of positive or negative market shifts in response to news events.

- **Understanding the Problem Statement**

- Global businesses face significant risks due to currency fluctuations resulting in losses, oftentimes.
- To manage these risks we need to predict currency movements and minimize the losses using best hedging strategies.
- We tried producing a predictive model that fetches historical currency data and uses it to predict future movements.
- For simulation, we have considered 4 popular Hedging Strategies viz. Butterfly, Straddle, and Collar.
- By comparing the standard key metrics for the strategies chosen by a user we recommend the best strategy to that user
- Further, we plan to notify users of significant currency movements and suggest immediate actions to mitigate risks

CURRENCY RISK ANALYSIS

- **Technologies Used**

- ARIMA model

- **Key Metrics**

- Sharpe Ratio: to measure risk-adjusted returns
- Max Drawdown: to measure the maximum loss

Key Features

- **Predictive Insights:**

- ARIMA model, forecasts future price movements.
- allowing it to test hedging strategies under predicted market conditions, enhancing decision-making with future trends.

- **Cumulative Performance Comparison :**

- Tool compares cumulative returns for all strategies
- Helping users see how each strategy performs over time and under varying market conditions

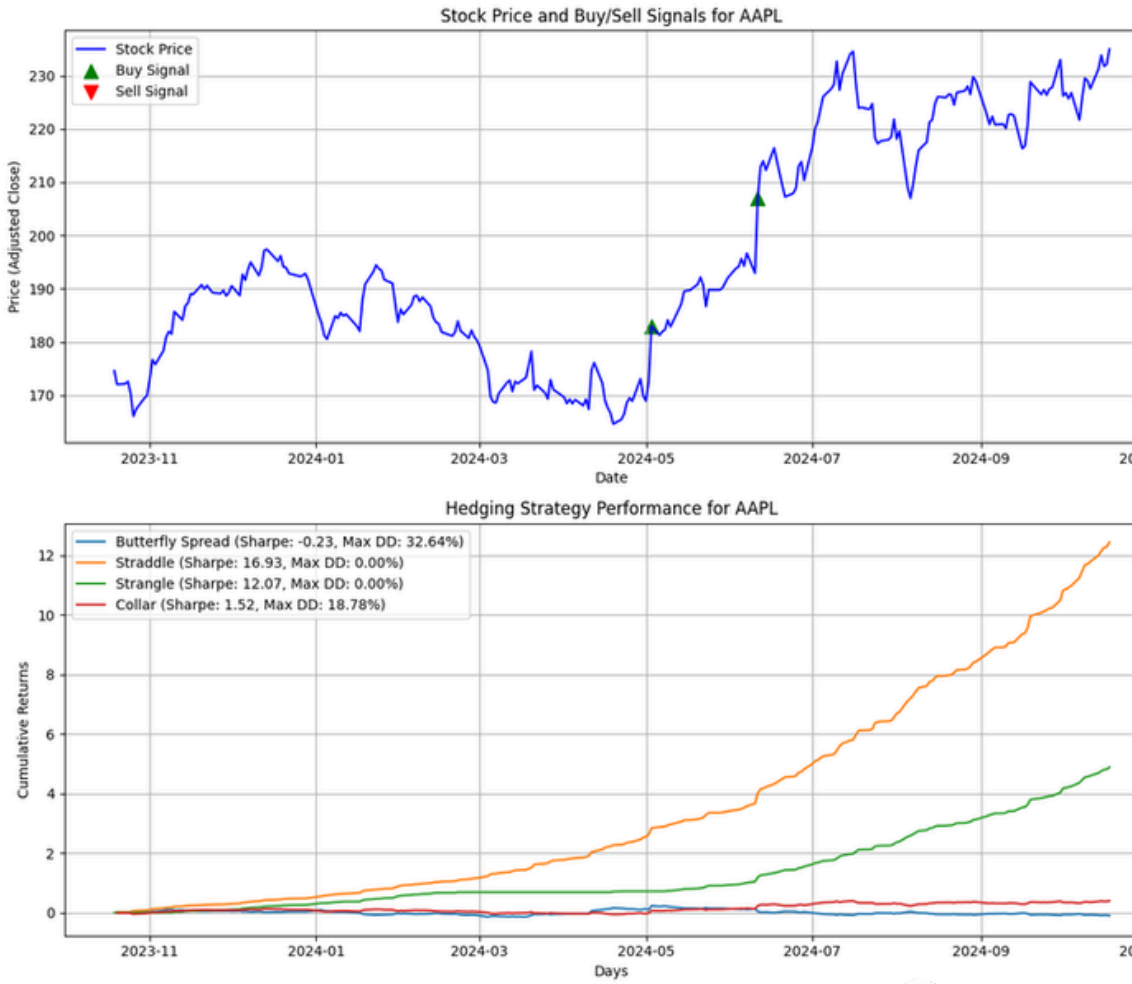
- **Currency Selection:**

- Choose a specific currency to analyze.
- Each currency represents its respective region (e.g., EUR for Europe, USD for the US).

Best Strategy Recommendation

Based on the performance metrics, we recommend Strategy - Butterfly.

Next



Save Report & Suggest Actions

Download Report

Suggested actions

Login / Signup

Email

Password

Login

or

Create Account

Enter Ticker Symbol

Euro

Euro

British Pound

Japanese Yen

Canadian Dollar

Australian Dollar

Swiss Franc

Chinese Yuan

Indian Rupee

Bitcoin

Ethereum

Litecoin

Ripple

Cardano

Binance Coin

Solana

Dogecoin

Polkadot

Chainlink

Next

User Workflow In Brief



Choose Hedging Strategies

Butterfly

Butterfly

Straddle

Collar

Next

- **Detailed user workflow for a fully functional tool***

1. Login/Signup: Enter credentials or create a new account.
2. Enter Ticker Symbol: Input the currency pair
3. Select Time Period: Choose the analysis period
4. Choose Hedging Strategies: Select the strategies to simulate (e.g., Butterfly, Straddle, Collar).
5. Input Optional Parameters: Customize strategy parameters if desired (e.g., strike prices, premiums).
6. Run Simulation: Click "Run Simulation" to analyze and compare strategies.
7. View Results: Review performance metrics and graphs for each strategy.
8. Best Strategy Recommendation: Check the system's recommended strategy based on performance.
9. Download or Save Report: Option to download results or save them to your account.
10. Start New Simulation: Repeat the process with different inputs or log out.

- **Conclusion and Opportunities:**

- We can combine solutions of both the sub-parts by integrating the tools together on the same portal/app , effectively contributing to the major features of the 'endgoal - dbPaint' application, such as
 1. Risk Management: By showing the reports and suggested actions
 2. Sentiment Analysis: By analyzing qualitative factors to give quantitative results
 3. Hyperconnected UI: By Integrating all the four clues on a single dashboard

THANK YOU

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